

Statistical dissociation of small carbon clusters: A phase space theory investigation

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Abstract

Unimolecular fragmentation in small neutral carbon clusters containing 4 or 5 atoms is studied by means of molecular dynamics (MD) simulations and the phase space theory (PST) statistical approach using various rigid body approximations. The clusters are modelled using the Takai–Lee–Halicioglu–Tiller potential, which predicts linear shapes as the most stable structures. The comparison of the distributions of kinetic energy and angular momentum released after dissociation shows that treating the fragments as linear may not be appropriate at high excitation energies. The branching ratios corresponding to the emissions of C₂ and C are also calculated.

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1. Introduction

Many intrinsic properties of atomic clusters are experimentally determined from the dissociation patterns of these molecules. Binding energies and relative stabilities are often estimated using a variety of statistical models, among which the Rice–Ramsperger–Kassel (RRK) and Weisskopf theories are probably the most famous but also the simplest. Actually, the application of statistical tools to cluster fragmentation should be considered carefully, fragmentation being an out-of-equilibrium process. This lead Klots to introduce the concept of the ‘evaporative ensemble’ [1].

An atomic cluster will be likely to dissociate one or more atoms depending on the time scale, the excitation

energy, but also the type of excitation. Upon photoabsorption, the system may reach a dissociative excited state and fragmentate before internal conversion had a chance to take place. This occurs for instance in the Ar₃⁺ cluster [2]. On the other hand, when the extra energy is brought via collision with an inert atom at a moderate speed, redistribution of energy is a relatively fast process with respect to the dissociation time scale.

In the seminal study by Weerasinghe and Amar [3], the phase space theory (PST) highly developed by Chesnavich and Bowers [4] following others [5] was seen to be accurate in reproducing MD results for the dissociation rate and the distribution of total energy released. Subsequently, Peslherbe and Hase [6] confirmed the general validity of PST for metal clusters. We have recently extended the work by Weerasinghe and Amar by investigating rotating atomic [7,8] and molecular [9] clusters, as well as heterogeneous systems [10]. One result of all these studies was the important role played by the

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rotational density of states (RDOS), a quantity which arises from the rigorous conservation of angular momentum during dissociation.

The subject of the present work, namely carbon clusters, is interesting in several respects. First, several experimental groups have studied their fragmentation pattern, arising either from photoabsorption (Choi et al. [11]) or collision (Martinet et al. [12]). Both these studies were carefully interpreted using statistical theories. The Neumark group [11] used parametrized PST and Monte Carlo (MC) simulations to quantify the extent of internal conversion in the photodissociation of linear C_n clusters, $n \leq 4 \leq 6$. The Wohrer and Chabot group [12] interpreted their high-energy collision-induced fragmentation experiments of C_n clusters, $n \leq 9$, with the help of the Weisskopf theory at the harmonic level built on sophisticated electronic structure calculations.

From a more methodological point of view, small carbon clusters are a challenge for phase space theory due to their linear ground state geometry [13], yet floppy at high energies. Within PST, the RDOS precisely depends on the rigid body approximations used for the fragments [4].

In the next Section, we briefly present the model and methods used to investigate thermal fragmentation in carbon clusters. Our results on the distributions of kinetic energy released (KER), final angular momentum and branching ratios are given and discussed in Section 3. We end in Section 4 by summarizing our results.

2. Numerical experiments

In order to achieve the significant amount of sampling required for the statistical treatment of fragmentation, we chose a relatively simple, explicit, potential energy surface for carbon clusters. The Takai–Lee–Halicioglu–Tiller (TLHT) potential [14] was seen to give much better results on small clusters [15] than the Brenner [16] or the Tersoff [17] potentials. The complete form and values of the parameters defining the TLHT potential can be found in the original Ref. [14] and they will not be recalled here. With this choice, the most stable structures of C_4 and C_5 are linear. The rhombohedral structure of C_4 and the ‘tadpole’ structure of C_5 are higher in energy than the linear structures by 0.93 eV and 1.26 eV, respectively.

We have performed constant-energy molecular dynamics simulations of cluster dissociation, at several total energies and zero angular momentum, using the Verlet algorithm with time step 0.25 fs. For both C_4 and C_5 , samples of 5×10^4 initial conditions were obtained from low-energy trajectories. After heating, each trajectory was followed until dissociation took place before 20 ps. After dissociation, we recorded the total

kinetic energy released, sum of its translational and rotational parts, as well as the final angular momentum and the branching ratio between atomic and diatomic fragmentation.

In addition to the MD trajectories, we have performed a statistical phase space theory analysis. For this we consider the C_n cluster at total energy E and angular momentum J . The final angular momentum after dissociation into the C_p and C_{n-p} fragments is denoted J_r , and L is the orbital momentum. The kinetic energy released is denoted ε_{tr} . From phase space theory [4] the unnormalized probability of finding a dissociation event with KER ε_{tr} and angular momentum J_r can be written as [3,8]

$$P(\varepsilon_{tr}, J_r, E, J) = \Omega_{p,n-p}(E - E_0 - \varepsilon_{tr}) \times \int_{L_{\min}}^{L_{\max}} \Gamma_{p,n-p}(\varepsilon_r^*, J_r) dL. \quad (1)$$

In the above equation, $\Omega_{p,n-p}$ is the total vibrational density of states (VDOS) of the product clusters, $\Gamma_{p,n-p}(\varepsilon_r^*, J_r)$ is the available volume of states with rotational energy lower than ε_r^* in angular momentum space. E_0 is the energy difference between the parent and product clusters (also functions of p and n). The boundaries $L_{\min}$ and $L_{\max}$ depend on both ε_{tr} and J_r , and must satisfy the laws of conservation of total energy and angular momentum. The KER distribution is obtained by integrating Eq. (1) over all possible values of J_r .

One assumption of phase space theory is that the transition state is located at the centrifugal barrier of the dissociation pathway. For a radial attractive potential of the form $-C/r^m$, $\varepsilon_r^* = \varepsilon_{tr} - \varepsilon^\dagger$, with $\varepsilon^\dagger$ the energy of the centrifugal barrier given by $L^{2m/(m-2)}/\Lambda$, Λ being a constant. Also, in PST the fragments are considered as rigid bodies. Thus the shapes of these rigid bodies are crucial in determining the expressions of the available sums and densities of rotational states.

After some algebra, one can easily derive the probability distributions of ε_{tr} in the case $J = 0$ [7,8]:

$$P(\varepsilon_{tr}, E, J = 0) \propto \Omega_{p,n-p}(E - E_0 - \varepsilon_{tr}) \tilde{\Gamma}_{p,n-p}(\varepsilon_{tr}), \quad (2)$$

with $\tilde{\Gamma}(\varepsilon_{tr})$ the rotational density of states under the constraint of zero total angular momentum

$$\tilde{\Gamma}_{p,n-p}(\varepsilon_{tr}) = \int_0^{J_r^{\max}} \Gamma_{p,n-p}(\varepsilon_r^*, J_r) dJ_r. \quad (3)$$

Similarly one can write the probability distribution of J_r as an integral of Eq. (1) over ε_{tr}

$$P(J_r, E, J = 0) \propto \int_{\varepsilon_{tr}^{\min}}^{E - E_0} \Omega_{p,n-p}(E - E_0 - \varepsilon_{tr}) \Gamma_{p,n-p}(\varepsilon_r^*, J_r) d\varepsilon_{tr}, \quad (4)$$

where $\varepsilon_{tr}^{\min}(J_r, E)$ is the minimum value of the KER at fixed J_r , and $\varepsilon_r^* = \varepsilon_{tr} - \varepsilon^\dagger(J_r)$.

In the present work, the fragmentation channels of small C_n clusters that we shall consider are the monomer and the dimer:



The largest fragment may or may not be treated as linear. Hence we will deal with the following rigid bodies approximations: (i) linear + atom, for (5); (ii) linear + linear, for (6); (iii) prolate top + atom, for (5); and (iv) sphere + linear, for (6) and $n = 5$.

Most expressions of the rotational sums of states corresponding to these four situations can be found in [18]. The configurational DOS's were obtained from parallel tempering MC simulations for clusters having 3 atoms or more, and by exact numerical integration for the dimer. From these quantities, the total vibrational DOS were obtained via a convolution product with the analytic kinetic DOS, $\Omega_k(E) \propto E^v$, v taking the value $n/2 - 1/2$ for linear fragments or $3n/2 - 4$ for tridimensional fragments. When the dissociation involves fragments larger than the monomer, one must account for the possible repartition of extra vibrational energy into the two fragments by convoluting the VDOS of one fragment with the sum of vibrational states of the other fragment (here, the dimer).

The finite temperature MC simulations also provided the average rotational constants of the product clusters, used as inputs in the calculations of the rotational DOS. The interaction between the fragments was obtained as an effective potential by randomly sampling the relative orientations at a fixed distance between centers of mass. The resulting energy was fitted as a $-C/r^p$ law, with m usually taking values in the range 5–8.

3. Results and discussion

Some typical KER distributions are represented in Fig. 1 for monomer and dimer dissociations. We find generally a very satisfactory agreement between MD and PST results, but a closer look at the predictions of the statistical theory tells us more about the relevance of the various approximations used.

In Fig. 1(a), PST was applied in the linear + linear assumption, with either harmonic or exact anharmonic vibrational DOS's. Clearly the harmonic approximation seems fully valid here, which is a little surprising due to the high internal energies involved. This indicates that, for the present dissociations, most extra energy goes into translational and rotational contributions.

The monomer emission from C_n can be described in the linear + atom or prolate top + atom approximations. For both $n = 4$ and $n = 5$ [see Fig. 1(b)], the two treatments yield quite similar KER distributions, mainly differing in their slope at $e_{tr} \rightarrow 0$. Here the MD data are

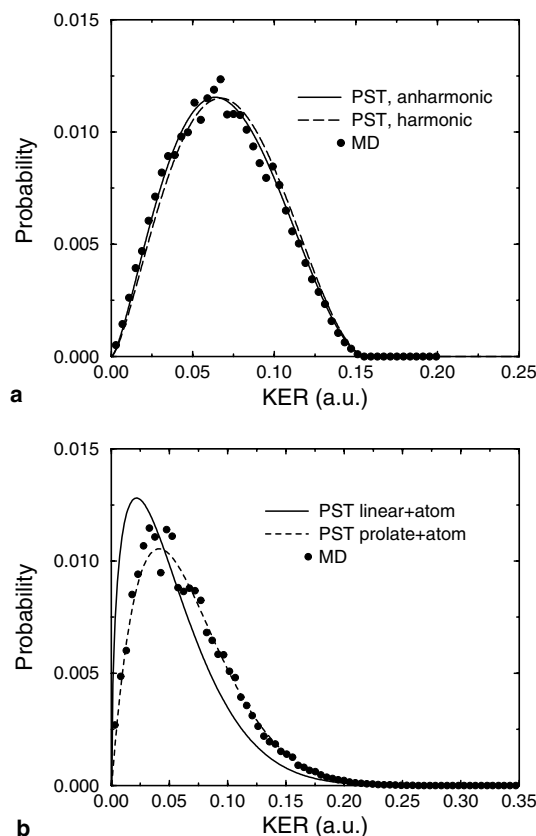


Fig. 1. Distributions of total kinetic energy released during the fragmentation of small C_n clusters. The results of molecular dynamics simulations are compared to predictions of phase space theory based on different rigid body approximations. (a) $C_4 \rightarrow C_2 + C_2$ with excitation energy 0.414 Hartree. (b) $C_5 \rightarrow C_4 + C$ with excitation energy 0.558 Hartree.

somewhat in between the two PST predictions, but significantly closer to the prolate top + atom results.

Further information is obtained through the distributions of angular momentum, plotted in Fig. 2 for the same conditions as Fig. 1. These distributions are more sensitive to the rotational sums and densities of states, therefore they are more likely to reflect the real effective shapes of the fragments. The effects of anharmonicities are only considered for the dissociation of C_4 into $2C_2$, in Fig. 2(a). Again, they are very weak and the MD data agree well with the PST results even at the harmonic level.

The fragmentation of C_2 from C_5 yields rather similar distributions of J_r for the two rigid body approximations used for C_3 (linear or spherical). However, only the latter is really consistent with MD in the whole range of angular momentum.

The linear + atom and prolate top + atom approximations produce very different distributions, due to the different expressions of the corresponding rotational sums of states [4]. This turns out to be a more stringent test of the PST approximations in Fig. 2(b), for the fragmentation of a monomer. The MD distributions

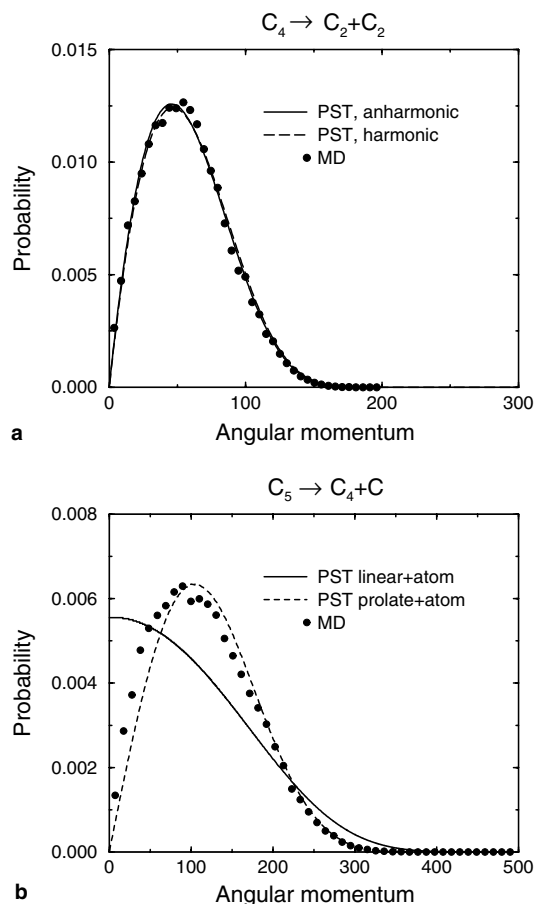


Fig. 2. Distributions of total angular momentum released during the fragmentation of small C_n clusters, in units of $\hbar$, for the same conditions as Fig. 1.

are much better reproduced using the prolate top approximation, especially at low angular momentum. Incidentally, accounting for the asymmetry of the tridimensional shapes does not prove to be essential, as the prolate top + atom results are indeed very close to PST calculations in the sphere + atom rigid body approximation. This shows that the main factor lies into the number of rotational degrees of freedom of the fragments.

The results above show that the largest fragment in the dissociation of small C_n clusters should not be treated as linear, even though the ground state geometry of the parent cluster is actually linear. The tridimensional character of C_3 and C_4 has two possible causes, namely bending in the linear shape and isomerization into other shapes. Since the total energies considered in Figs. 1 and 2 are much above the isomerization energies of C_3 and C_4 , it is reasonable to consider these isomers as effectively tridimensional. This can be verified by examining the variations with temperature of the largest rotational constants of these clusters, which roughly display a $1/T$ behavior.

Another experimentally relevant quantity is the branching ratio between the two fragmentation channels (5) and (6). At a given total energy, the relative probabilities of emitting in different channels only depend on the fragments properties [3], through integrals of Eq. (2) over ε_{tr} . We determine the normalizing constant by fitting the PST value to the exact MD result [10]. We have represented in Fig. 3 the variations of the probability of emitting a dimer rather than a monomer, for both the C_4 and C_5 parents, using two approximations of PST where the largest fragments are described either as linear or as tridimensional.

For both C_4 and C_5 dissociations, the nonlinear rigid bodies approximations are clearly better at high excitation energies. Extra simulations have been performed at moderate excitations, namely 0.31 Hartree above the ground state. At these energies, the branching ratios are closer to the linear approximation for the fragments. This is consistent with the expected greater rigidity of the products. However, it probably points out a limitation of the statistical approach for these systems exhibiting a variable degree of floppiness.

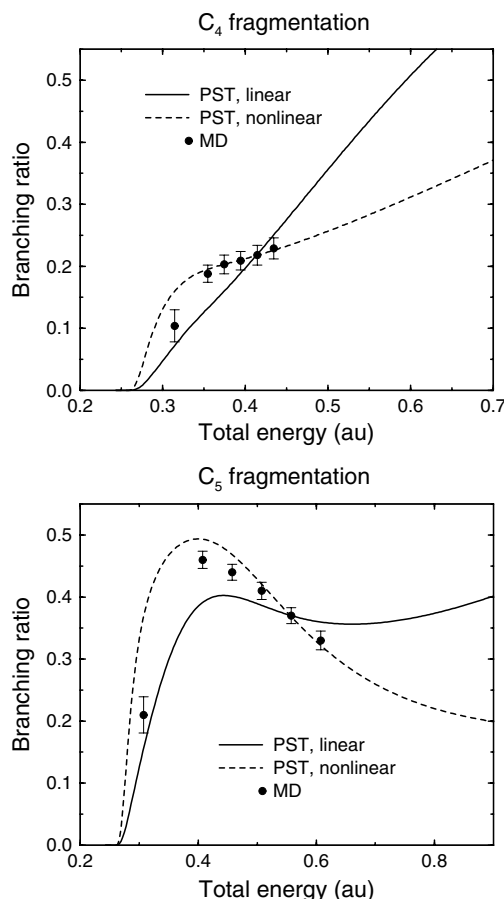


Fig. 3. Probability of dissociation of C_n into the $C_{n-2} + C_2$ channel, with respect to the $C_{n-1} + C$ channel, versus total excitation energy. The results of molecular dynamics simulations are compared to predictions of phase space theory based on different rigid body approximations for C_{n-1} and C_{n-2} .

Unfortunately, the amount of data to which we can compare our own results is rather limited. Still, the branching ratios represented in Fig. 3 follow similar variations as the experimental and Weisskopf results given by Martinet et al. [12]. In particular, the probability that C_5 dissociates into $C_2 + C_3$ has a maximum. The quantitative values are however quite different from the Weisskopf analysis, obviously due to the use of a much simpler model for carbon clusters. The relatively good agreement partly comes from our observation that the largest fragments are better described as spherical than linear, an approximation included in the Weisskopf approach under the form of a rotational DOS proportional to ε_{tr} . The agreement also probably results from the overall good performance of the harmonic approximation for the VDOS, also used in the calculations of Ref. [12]. Interestingly, the nonmonotonic variations of the branching ratio for C_5 have also been obtained in multi-fragmentation Monte Carlo simulations [19] in the Gross and Hervieux model [20].

The major drawback of the present work probably remains the use of an oversimplified atomistic potential. Yet, before similar investigations are carried out with more sophisticated models, some methodological improvements to the PST approaches should be undertaken. Accounting for the presence of different isomers with very different shapes seems here the main issue. It could be addressed with a superposition approximation to the VDOS's [21,22], along with a master equation description of dissociation.

4. Conclusion

Fragmentation in small carbon clusters was investigated by employing a statistical approach based on the phase space theory developed by Chesnavich and Bowers [4] among others. To assess quantitatively this approach and to fulfill the need of a large number of molecular dynamics trajectories, we had to select a simple potential energy surface, which still captures most of the structural and vibrational features of these clusters [14,15].

Obviously this potential is not able to account for the complex electronic redistribution processes taking place during the fast excitation by photoabsorption or by energetic collision. From this point of view the present approach could be seen as complementary to that of Martinet et al. [12], where accurate electronic structure calculations were used in simpler statistical frameworks.

The general validity of the PST approach, as it stands here, shows that accurate predictions of experimental observables will be conceivable with much better atomistic models in the near future.

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